

APPFREENSN Process Module Map - DD Aligned Update

Version: 2026-06-01

Source PDFs updated:

- APPFREESN-PROCESS_MODULE_MAP-120526-124342.pdf
- Finalized DD corpus under 99_rewritten
- Performance/complexity assessment:
99_analysis/Sophix_V3_Performance_Complexity_Assessment_20260601.md
- MVP runtime simplification baseline:
99_analysis/Sophix_V3_MVP_Runtime_Simplification_Baseline_20260601.md

1. Purpose

This document updates the older process-to-module map so it matches the finalized SOPHIX DD set.

The map answers four questions for implementers:

1. Which module owns the process?
2. Which modules participate?
3. Which runtime mechanisms are used?
4. Which DD should a developer open for implementation details?

Owner means the module that owns the authoritative state or drives the workflow. Participant means a module that provides an API, event consumer, worker, user task, reference data, or external side effect.

2. Reading Rules

- A DD file is not automatically a separate pod or service.
- A worker-topic name is not a deployment name. It is a Camunda external-task topic.
- One worker app may handle many low-volume topics if metrics and retries remain visible per topic.
- Commands use REST/internal APIs. Facts use Kafka events.
- Source-of-truth reads come from owning APIs or approved client wrappers, not cross-module SQL.
- Redis/read-through cache can speed reference reads, but command flows must be able to fall back to the source API.
- Modules do not communicate all-to-all. Each process has one owner; participants are called only when the process needs their authoritative data or side effect.
- Reporting dashboards and exports use REP-01, not live fan-out to operational modules.
- Dense frontend read screens may use a BFF/read aggregator, but the aggregator must not own business state.

3. Foundation Map

DD	Owner	Used by	Runtime role
FOUNDATION_AUTH	Platform/Auth	All modules	Keycloak/OIDC, JWT validation, service tokens, role model.
FOUNDATION_DB	Platform/Data	All modules	Three PostgreSQL clusters, PgBouncer, replication, backup, partitioning rules.
FOUNDATION_KAFKA	Platform/Eventing	All modules with events	Kafka topics, envelope, outbox/inbox, replay and retry.
FOUNDATION_CACHE	Platform/Cache	Modules reading reference data	Redis/Valkey non-authoritative cache and invalidation rules.
FOUNDATION_CAMUNDA	Platform/Workflow	FUL, SUB-WF, WO, OSR, BIL	BPMN deployment, external tasks, user tasks, timers, message correlation.

DD	Owner	Used by	Runtime role
FOUNDATION_DROOLS	Platform/Rules	Modules with policy rules	KIE/Drools deployment and rule execution contract.
FOUNDATION_FILE_STORAGE	Platform/Storage	BIL, ILM, WO, OSR, NOT	S3/MinIO object storage for PDFs, KYC docs, evidence, attachments.

4. Catalog And Configuration Processes

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_PLM-CFG-01 Service Catalog	PLM/catalog	BIL, FUL, SUB-LM, search/cache consumers	REST, Kafka, cache, optional Drools	Service definitions and lifecycle.
DD_SIP-01 Package Management	SIP/package catalog	SUB-LM, BIL, FUL, RLM	REST, Kafka, cache, Drools	Package/product offer definitions and package lookups.
DD_RLM-CFG-01 HomePass Configuration	RLM/HomePass	SUB-LM, FUL, WO, ILM	REST, Kafka, cache	HomePass, coverage, topology, eligibility reference data.
DD_ILM-CFG-02 Tech Region Registry	ILM/config	WO, FUL, EM	REST, cache, Drools where configured	Tech regions, candidate groups, dispatch references.
DD_PLM-CFG-02 Tax Configuration	PLM/BIL config	BIL-02-TAX, BIL-01, BIL-02	REST, cache, Drools	Tax rules and operator tax setup.
DD_PLM-CFG-03 Wallet Configuration	PLM/BIL config	BIL-05, BIL-01	REST, cache, Drools	Wallet behavior, channels, limits.
DD_PLM-CFG-04 Discount Catalog	PLM/BIL config	BIL-01, BIL-02, SIP	REST, cache, Drools	Discount definitions and eligibility.
DD_PLM-CFG-05 Adjustment Type Catalog	PLM/BIL config	BIL-02-ADJ	REST, cache	Adjustment type definitions.
DD_PLM-CFG-08 Adjustment Catalog	PLM/BIL config	BIL-02-ADJ	REST, cache	Adjustment reason codes and defaults.
DD_PLM-CFG-06 Equipment Type Catalog	PLM/OSR config	OSR, WO, FUL	REST, cache	Equipment kinds, SKUs, compatibility, fees.
DD_BIL-CFG-01 Billable Event Catalog	BIL config	BIL-01, FUL, SUB-WF	REST, Kafka/cache	Events that can become charges.

5. Customer, Workforce, Notification, And Communication Processes

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_ILM-CFG-01 Customer Service	ILM Customer Service	SUB-LM, FUL, BIL, WO, NOT	REST, Kafka, Camunda user tasks where KYC applies	Customer, account, KYC, contacts, preferences, search, notes.
DD_TCK-01 Ticketing and Case Management	TCK	ILM, SUB-LM, SUB-WF, BIL, FUL, WO, OSR, NOT/ICN, file storage	REST, Kafka/outbox/inbox, optional Camunda where future case workflows need it	Tickets/cases, SLA, assignment, timeline, attachments, linked entities, ticket-to-WO creation.
DD_EM-02 Contractor and Staff Registry	EM	WO, FUL, NOT, ICN, OSR	REST, cache	Canonical contractor/staff/capacity master data. Supersedes the older contractor team/agent registry.
DD_NOT-01 Notification Service	NOT	All workflow modules, external channel providers	REST, Kafka, file storage, provider adapters	Template, dispatch, retries, delivery status.
DD_ICN-01 Internal Communications	ICN	FUL, WO, ILM, BIL, NOT	REST, Kafka	Staff notification inbox, ack, internal messages.
DD_EM-CFG-03 RBAC Catalog	EM config	All frontends and module APIs	REST/cache, auth integration	Business roles, permissions, scopes, UI visibility mapping.
DD_EM-CFG-04 Approval Workflow Catalog	EM config	Billing, OSR, commercial, ASR, field audit	REST, Camunda user tasks where needed	Reusable approval definitions; modules still own final business state.

6. Subscription Core And Lifecycle

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_SUB-LM-01 Subscription Lifecycle Management	SUB-LM	ILM, SIP, RLM, SUB-WF, BIL, FUL	REST, Kafka/outbox	Subscription source of truth. No workflow orchestration here.
DD_SUB-WF-FRAMEWORK-01 Subscription Workflow Framework	SUB-WF	Camunda, SUB-LM, satellite flows	REST, Camunda, Kafka/outbox	Operation ledger, idempotency, concurrency, cancellation, process start.
DD_SUB-WF-FRAMEWORK-01-WORKERS	SUB-WF	BIL, FUL, OSR, NOT, SUB-LM	Camunda external tasks, REST	Shared worker-topic contracts.
DD_SUB-WF-ACTIVATE-01 Subscription Activation	SUB-WF activation	SUB-LM, FUL, BIL, OSR, NOT	REST, Drools, Camunda, Kafka	Activate an existing created subscription.
DD_SUB-WF-ACTIVATE-01-FLOW-WIK	SUB-WF activation	WIK-specific BIL/FUL/OSR/NOT dependencies	Drools, Camunda	WIK activation BPMN/rules composition.
DD_SUB-WF-PAUSE-01 Subscription Pause	SUB-WF pause	SUB-LM, BIL, FUL, NOT	REST, Drools, Camunda, Kafka	Customer/operator pause operation.
DD_SUB-WF-RESUME-01 Subscription Resume	SUB-WF resume	SUB-LM, BIL, FUL, NOT	REST, Drools, Camunda, Kafka	Resume from paused/restricted/suspended states.
DD_SUB-WF-TERMINATE-01 Subscription Termination	SUB-WF terminate	SUB-LM, BIL, FUL, OSR-RMA, NOT	REST, Drools, Camunda, Kafka	Termination, finalization, equipment disposition.

7. Subscription MACD And Operational Workflows

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_SUB-WF-UPGRADE-01 Subscription Upgrade	SUB-WF upgrade	SUB-LM, SIP/PLM, BIL, FUL, WO, OSR, NOT	REST, Drools, Camunda, Kafka	Upgrade package/service with price and equipment impact.
DD_SUB-WF-DOWNGRADE-01 Subscription Downgrade	SUB-WF downgrade	SUB-LM, SIP/PLM, BIL, FUL, NOT	REST, Drools, Camunda, Kafka	Downgrade package/service with billing impact.
DD_SUB-WF-RESTRICT-01 Subscription Restriction	SUB-WF restrict	SUB-LM, BIL, FUL, NOT	REST, Drools, Camunda, Kafka	Commercial or policy restriction.
DD_SUB-WF-SUSPEND-NP-01 Non Payment Suspension	SUB-WF suspend NP	BIL-04, SUB-LM, FUL, NOT	REST, Drools, Camunda, Kafka	Dunning-triggered suspension for non-payment.
DD_SUB-WF-RELOCATION-01 Subscription Relocation	SUB-WF relocation	RLM, SUB-LM, FUL, WO, OSR, BIL, NOT	REST, Drools, Camunda, Kafka	Move service to another HomePass/address.
DD_SUB-WF-MIGRATION-01 Subscription Migration	SUB-WF migration	RLM, SIP, SUB-LM, FUL-10, WO, OSR, BIL, PROV-INT, NOT	REST, Drools, Camunda, Kafka	Package/HomePass migration. Cross-technology technical fulfillment is delegated to FUL-10.

Each process above also has a WIK flow DD. The parent DD defines the generic contract. The WIK flow DD defines operator-specific BPMN, Drools, sample payloads, and runbook behavior.

8. Fulfillment And Order Capture

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_FUL-02-FRAMEWORK	FUL order framework	ILM, SIP/RLM, BIL, WO, SUB-LM, SUB-WF	Camunda, REST, Kafka	Shared order state, workers, message catches, cancellation.
DD_FUL-02-Order Capture	FUL	ILM, SIP/RLM, BIL, WO, SUB-LM, SUB-WF, NOT	Camunda, REST, Kafka, Drools where declared	New-customer onboarding and order orchestration.
DD_FUL-02-STEP-CAPTURE	FUL	ILM, RLM/SIP	REST, Camunda worker	Capture/validate initial order request.
DD_FUL-02-STEP-VALIDATE	FUL	ILM, SIP/RLM, BIL	REST, Drools, Camunda worker	Eligibility, package, HomePass, account checks.
DD_FUL-02-STEP-KYC	FUL/ILM	ILM, NOT/ICN	Camunda user task, REST	KYC approval path and notifications.

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_FUL-02-STEP-PAYMENT	FUL/BIL	BIL-01-PAY, NOT	Kafka message catch, REST	Expected payment, payment wait, callbacks.
DD_FUL-02-STEP-INSTALL	FUL/WO	WO, EM, OSR, NOT	Camunda worker, message catch	Create installation work order and wait for finalization.
DD_FUL-02-STEP-SUBSCRIPTION	FUL/SUB-LM	SUB-LM, ILM	REST, Camunda worker	Create subscription after prerequisites.
DD_FUL-02-STEP-ACTIVATION	FUL/SUB-WF	SUB-WF activation, BIL, OSR	REST, Camunda worker	Trigger activation from order flow.
DD_FUL-02-STEP-CANCELLATION	FUL	BIL, WO, SUB-LM, NOT	Camunda compensation, Kafka	Cancel and compensate order.
DD_FUL-03 Service Activation	FUL	SUB-WF, BIL, OSR, provisioning adapter	REST, Camunda, Kafka	Activate service on network and report result.
DD_FUL-04 Restriction Fulfillment	FUL	SUB-WF/BIL-04, provisioning adapter, NOT	REST, Camunda, Kafka	Apply/remove network restriction.
DD_FUL-05 Termination Fulfillment	FUL	SUB-WF, BIL, OSR-RMA, provisioning adapter	REST, Camunda, Kafka	Deactivate service and trigger equipment disposition.
DD_FUL-07 HomePass Lifecycle Fulfillment	FUL/RLM	RLM, WO, NOT	REST, Camunda/Kafka where declared	HomePass lifecycle side effects.
DD_FUL-08 Billing Anchor Date Change	FUL/SUB-WF	SUB-LM, BIL	REST, Camunda where needed, Kafka	Controlled billing-anchor change execution support.
DD_FUL-09 Equipment Swap Fulfillment	FUL	OSR-RMA, WO, PROV-INT, BIL	REST, Kafka, Camunda where needed	Service/network rebinding side of equipment swap. OSR remains stock owner.
DD_FUL-10 Technology Migration Fulfillment	FUL	SUB-WF migration, WO, OSR, FUL-09, PROV-INT, RLM	REST, Kafka, Camunda where needed	Technical cross-technology migration sequence.
DD_PROV-INT-01 Provisioning Integration	FUL/integration	FUL-03/04/05/09/10, SUB-WF, external OSS	REST/adapters, Kafka, scheduled reconciliation	BSS-side provisioning dispatch, retry, force-sync, reconciliation.

9. Work Order Processes

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_WO-01 Work Order Module	WO	FUL, SUB-WF, OSR, ILM, EM	REST, Kafka, Camunda workers	Work order source of truth and parent contract.
DD_WO-01-FRAMEWORK	WO	EM, ILM, OSR, NOT	REST, Kafka, Camunda	State machine, assignment, claim, finalization, checklist.
DD_WO-01-FLOW-INSTALLATION	WO	FUL, RLM, OSR, EM, NOT	Camunda, REST, Kafka, file storage	Installation work orders, equipment binding, topology confirmation.
DD_WO-01-FLOW-SHIFTING	WO	SUB-WF relocation, RLM, OSR, EM, NOT	Camunda, REST, Kafka	Disconnect/reconnect phases for shifting.
DD_WO-01-FLOW-SUPPORT	WO	ILM, OSR, EM, NOT	Camunda, REST, Kafka	Support visits, desk-resolved branch, escalation.
DD_WO-01-CONTRACT-FOR-SUB-WF	WO/SUB-WF	SUB-WF relocation/migration/termination	REST, Kafka	Contract used by subscription workflows to request/observe WOs.

Ticketing integration note: TCK creates support/follow-up Work Orders through WO-01 POST `/api/work-orders` when a ticket category allows field intervention. WO remains the owner of field execution. TCK consumes WO events and updates the ticket timeline/status.

10. OSR And Equipment Processes

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_OSR-01 Stock Chain Engine	OSR	WO, FUL, EM	REST, Kafka	Stock locations, movements, reservations, balances.
DD_OSR-INSTANCE-01 Equipment Instance Registry	OSR	WO, FUL, SUB-WF	REST, Kafka	Equipment instance lifecycle and serial authority.

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_OSR-RMA-01-FRAMEWORK	OSR-RMA	WO, EM, FUL, BIL	Camunda, REST, Kafka	Common swap/RMA state machine and worker contracts.
DD_OSR-RMA-01 Equipment Swap and RMA	OSR-RMA	WO, BIL, NOT	Camunda, REST, Kafka	Swap request parent contract.
DD_OSR-RMA-01-FLOW-SWAPP-GPON	OSR-RMA	WO, FUL, OSR instance, NOT	Camunda, REST, Kafka	GPON swap flow.
DD_OSR-RMA-01-FLOW-SWAPP-HFC	OSR-RMA	WO, FUL, OSR instance, NOT	Camunda, REST, Kafka	HFC swap flow.
DD_OSR-RMA-01-FLOW-EQP	OSR-RMA	WO, OSR stock/instance, NOT	Camunda, REST, Kafka	Equipment pickup/recovery flow.
DD_OSR-RMA-01-FLOW-EQU	OSR-RMA	WO, OSR stock/instance, NOT	Camunda, REST, Kafka	Equipment upgrade/exchange flow.
DD_OSR-02 Procurement	OSR procurement	EM-CFG-04, suppliers, OSR stock	REST, approvals, Kafka	Purchase requests/orders, receiving, stock update.
DD_OSR-05 Inventory Audit	OSR audit	EM-CFG-04, WO/field audit, OSR stock/instance	REST, approvals, scheduled jobs, Kafka	Physical counts, discrepancies, write-off/adjustment flow.

11. Billing Processes

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_BIL-01 Charging Engine Core	BIL charging	SUB-LM, FUL, PLM/SIP, ILM	REST, Kafka, Drools	Charge computation and billable events.
DD_BIL-01-PAY-01 Payment Application	BIL payment	Payment providers, FUL-02, SUB-WF, NOT	REST callbacks, Kafka, Drools	Payment receive, classify, apply, reverse.
DD_BIL-01-CN-01 Note Application	BIL notes	ILM, NOT, file storage	REST, Kafka	Credit/debit note behavior.
DD_BIL-02 Invoicing Core	BIL invoice	ILM, SUB-LM, FUL, PLM/SIP, file storage	REST, Kafka	Invoice model and lifecycle.
DD_BIL-02-GEN-01 Invoice Generation	BIL generation	SUB-LM, FUL, PLM/SIP, ILM	Scheduled job/Camunda, Kafka	Pro-forma/final invoice generation.
DD_BIL-02-READ-01 Invoice Read API	BIL read	ILM, customer/backoffice UI	REST, file storage	Invoice query/download APIs.
DD_BIL-02-TAX-01 Tax Invoice and Gateway	BIL tax	Tax authority adapter, file storage	REST, Kafka, Drools	Tax invoice signing, retry, cancellation.
DD_BIL-02-STA-01 Status Management and Recovery	BIL invoice status	BIL operations	REST, scheduled recovery	Invoice status transitions and repair.
DD_BIL-02-ADJ-01 Invoice Adjustments	BIL adjustments	PLM adjustment catalogs, ILM, NOT	REST, Drools, Kafka	Manual/system adjustments and approvals.
DD_BIL-03 Cycle Close	BIL cycle close	SUB-LM, BIL-02, BIL-01, NOT	Scheduled scanner, Camunda/Kafka	Cycle evaluation, close, failure recovery.
DD_BIL-04 Dunning Engine	BIL dunning	SUB-WF suspend/restrict/resume, NOT, ILM	Scheduled scanner, Drools, Camunda, Kafka	Debt escalation and non-payment actions.
DD_BIL-05 Wallet and Topup	BIL wallet	Payment app, ILM, NOT	REST, Kafka	Wallet ledger, balance, aliases, top-up channels.
DD_BIL-CFG-01 Billable Event Catalog	BIL config	BIL-01, FUL, SUB-WF	REST, cache/Kafka	Chargeable event definitions.
DD_PAY-GW-01 Payment Gateway Integration	BIL/payment integration	Payment providers, BIL payment, FUL-02, SUB-WF, NOT	REST callbacks, Kafka, reconciliation jobs	Gateway callbacks, idempotency, settlement, repair.
DD_MED-01 Usage Mediation	BIL usage	VoipSwitch/CDR source, RAT-01, BIL wallet	Batch/stream ingestion, quarantine, Kafka	Offline CDR collection, normalization, deduplication.
DD_RAT-01 Offline Rating	BIL rating	MED-01, PLM tariff catalog, BIL wallet	Batch jobs, Drools/tables where configured, Kafka	Offline usage rating and wallet debit integration.

11A. Commercial, Assurance, Reporting, And Channel Processes

Process/DD	Owner	Participants	Runtime mechanisms	Developer focus
DD_SALES-01 Lead Territory and Franchise Sales Management	SALES	EM-01, RLM, ILM, FUL-02, REP-01	REST, Kafka, cache	Leads, field activity, territory, conversion attribution.
DD_EM-01 Franchise Management	EM/franchise	SALES, RLM, FUL, REP-01	REST, Kafka/cache	Franchise registry, lifecycle, commercial scope.
DD_SIP-03 Discount Assignment	SIP/commercial	SIP-05, DIS-OP-01, BIL, EM-CFG-04	REST, Drools, approvals, Kafka	Customer/subscription discount assignment and cancellation.
DD_SIP-04 Bundle Launch	SIP/catalog	PLM, BIL, SALES, FUL	REST, approvals, Kafka/cache	Bundle/package launch lifecycle.
DD_SIP-05 Promotional Campaigns	SIP/commercial	SALES, DIS-OP-01, BIL, CVM	REST, Drools, approvals, Kafka	Campaign setup, eligibility, launch, retirement.
DD_DIS-OP-01 Discount Runtime Application	DIS/BIL support	BIL-01, SIP-03, SIP-05	REST/internal rules, Drools/tables	Billing-time discount application.
DD_TCK-01 Ticketing and Case Management	TCK	ILM, SUB, BIL, WO, OSR, NOT/ICN	REST, Kafka, file storage	Ticket engine, SLA, assignment, timeline, ticket-to-WO.
DD_ASR-01/02/03/04 Service Assurance Flows	ASR/TCK satellite	TCK, ILM, SUB, BIL, WO, OSR, NOT/ICN, EM-CFG-04	REST, Camunda where needed, Drools where configurable	Technical trouble, information request, complaint, service request flows on TCK.
DD_EM-03 CVM Retention	EM/CVM	ILM, SUB, BIL, TCK, SIP/DIS, NOT, REP-01	Scheduled jobs, REST, Kafka, rules	Retention queues, recovery actions, outcome tracking.
DD_FA-01/02/03 Field Audits	Field audit capability	WO, OSR, RLM, ILM, EM-02, file storage, TCK	REST, WO, Kafka, approvals	Network, customer premises, equipment/stock audit processes.
DD_CUST-INT-01 Customer Interaction Timeline	Customer timeline/read model	ILM, TCK, SALES, CVM, NOT, BIL/SUB references	Kafka consumers, REST read API	Cross-module customer interaction view.
DD_REP-01 Reporting Data Mart and Dashboard API	REP	All event-producing modules	Kafka/event ingestion, scheduled extracts, REST/export APIs	Dashboards, exports, reconciliation, management reporting.
FE-APP-00 to FE-APP-05	Frontend suite	Module APIs through gateway, REP-01 for reports	Web/PWA/mobile, auth, API gateway	Shared frontend foundation, backoffice, sales, contractor, self-care, reporting.
FE-CH-USSD-01 USSD Channel	USSD adapter	ILM, SUB, BIL/PAY-GW, TCK, NOT	Telco callbacks, session store, REST APIs	Feature-phone self-care channel without duplicating business state.

12. Cross-Process Event Map

Event area	Typical producers	Typical consumers
Customer/account events	ILM Customer Service	TCK, BIL, SUB-LM, FUL, WO, NOT, search/reporting.
Ticket events	TCK	ILM/Customer 360, WO, NOT/ICN, reporting, audit.
Catalog/config events	PLM/SIP/RLM/BIL-CFG/ILM-CFG	SUB-LM, FUL, BIL, WO, OSR, cache consumers.
Subscription events	SUB-LM and SUB-WF satellite flows	BIL, FUL, ILM, NOT, reporting, cache consumers.
Billing events	BIL charging/payment/invoice/dunning/wallet	SUB-WF, FUL, NOT, ILM/customer views, reporting.
Fulfillment events	FUL-02/03/04/05/07	SUB-WF, BIL, WO, OSR, NOT.
Work order events	WO	TCK, FUL, SUB-WF, OSR, ILM, EM, NOT.
Equipment events	OSR	WO, FUL, SUB-WF, BIL, reporting.
Procurement/audit events	OSR-02/05, FA-01/02/03	OSR, WO, EM-CFG-04, REP-01, audit consumers.
Payment gateway events	PAY-GW-01/BIL payment	BIL ledger, FUL-02, SUB-WF, NOT, REP-01.
Usage mediation/rating events	MED-01/RAT-01	BIL wallet, REP-01, customer usage views where allowed.
Commercial/CVM events	SALES, SIP, DIS-OP, EM-03	FUL, BIL, NOT, REP-01, customer timeline.
Notification events	NOT/ICN	Originating modules, audit/reporting.

12A. Performance Map

This process map has many modules, but the runtime must avoid all-to-all synchronous calls.

Pattern	Allowed	Not allowed
Command ownership	One module owns each write command and state transition.	A frontend or BFF writing multiple module states directly.
Supporting reads	Read from owning APIs, cache, or documented read models.	Cross-module SQL joins from another module database.
Dashboards	REP-01 aggregates and serves dashboards/exports.	Dashboard screens calling every operational module for totals.
Customer 360	Panelized reads with timeouts; optional BFF/read aggregator.	One blocking chain where failure of one module breaks the whole screen.
Cache	Reference/catalog/session acceleration.	Final truth for payments, balances, subscription status, work order finalization, equipment ownership, or ticket status.
Workers	Consolidated worker apps with per-topic metrics.	One mandatory pod per worker topic.
Camunda	Long-running workflows, timers, user tasks, compensation.	Simple CRUD, basic validation, read-only reporting aggregation.
Drools	Operator-configurable business policy.	Fixed field/enum/FK/idempotency validation.

13. Implementation Sequence

Recommended dependency order:

1. Foundations.
2. ILM Customer Service and catalog/config capabilities.
3. SUB-LM.
4. SUB-WF framework and worker catalog.
5. Billing payment/charging/invoice basics.
6. WO framework and EM registry.
7. OSR stock/equipment instance basics.
8. FUL-02 order capture and FUL service execution flows.
9. Subscription lifecycle flows.
10. Subscription MACD flows.
11. TCK Ticketing before final Backoffice CRM/ticketing delivery, because support cases may create WOs and must appear in Customer 360.
12. Billing cycle close, dunning, wallet, tax recovery, operational refinements.
13. NOT/ICN integrations should be started early and hardened throughout, because every workflow eventually depends on notification.
14. PAY-GW, PROV-INT, MED/RAT, OSR procurement/audit, ASR flows, CVM, REP-01, and USSD according to the first implementation wave.

14. What To Ignore From The Old Map

The older process map should not be used for:

- assigning customer ownership to billing;
- treating onboarding, field-ops, inventory, or homepass as the final module boundaries;
- treating one worker topic as one deployment;
- adding business logic into gateway/adapters;
- bypassing ILM, SUB-LM, BIL, WO, OSR, or catalog APIs with cross-database SQL.

Where this document conflicts with the old PDF, this DD-aligned version is the current reference.